Term review and consolidation

Lecture and worked examples

Individual review

Work without the course notes for 25 minutes, then check with the methods in the booklet. For each question, note: solved alone, solved with help, or to redo. Justify the answers.

Individual review

1. Reduce $90/126$ to lowest terms. Prove that the sum of two odd integers is even.
2. With $A(-1,2)$ and $B(3,-2)$, calculate $\overrightarrow{AB}$, AB and the midpoint of $[AB]$ in an orthonormal frame.
3. Determine an equation of (AB); does $C(5,-4)$ belong to this line?
4. Calculate $[-2,4] \cap (1,7]$ and $[-2,4] \cup (1,7]$.
5. Solve $|x-1| \leq 4$ and explain why $1/3 \notin \mathbb{D}$.

Worked answer: Individual review

1. $\gcd(90, 126) = 18$, so $90/126 = 5/7$. If $u = 2k + 1$ and $v = 2\ell + 1$, then $u + v = 2(k + \ell + 1)$ is even.
2. $\overrightarrow{AB} = (4, -4)$; $AB = 4\sqrt{2}$; midpoint $I(1, 0)$.
3. $y = -x + 1$. For C, $-4 = -5 + 1$: it belongs to (AB).
4. Intersection $(1, 4]$; union $[-2, 7]$.
5. $x \in [-3, 5]$. Writing $1/3 = a/10^n$ would require 10^n to be divisible by 3, which is impossible.

Correcting an error and revisiting a method

Correct: (a) "$\overrightarrow{AB} = (x_A - x_B, y_A - y_B)$"; (b) "The zero vector directs every line"; (c) "$|x - 2| \leq 3$ gives $x \in [2, 3]$"; (d) "$1/3 \approx 0.333$, so $1/3$ is decimal".

Worked answer: Correcting an error and revisiting a method

(a) Subtract the origin from the endpoint: $(x_B - x_A, y_B - y_A)$. (b) A direction vector must be nonzero. (c) The distance to 2 is at most 3: $x \in [-1, 5]$. (d) 0.333 differs from $1/3$: $3 \times 0.333 = 0.999 \neq 1$. A decimal approximation does not determine the nature of the approximated number.

Correcting an error and revisiting a method

1. Reduce $72/108$, then $\dfrac{3}{4} - \dfrac{1}{6}$.
2. With $P(1,1)$ and $Q(5,3)$, determine the midpoint, a direction vector and an equation of (PQ).
3. Determine $(-\infty, 3] \cap [-1, +\infty)$, then solve $|x+2| < 1$.

Worked answer: Correcting an error and revisiting a method

1. $2/3$; $7/12$. 2. Midpoint $(3, 2)$, direction vector $(2, 1)$, equation $x - 2y + 1 = 0$. 3. Intersection $[-1, 3]$; solution of the inequality $(-3, -1)$.

Applying and self-assessing

Two displacements are represented by $\vec{u}(6,4)$ and $\vec{v}(9,6)$.

1. Show that they are collinear and find the coefficient relating $\vec{v}$ to $\vec{u}$.
2. The starting point is $A(-2,1)$. Determine the endpoints B and C associated with the displacements $\vec{u}$ and $\vec{v}$.
3. Find an equation of the line through these three points.
4. A point $M(x,y)$ of this line must satisfy $|x-4| \leq 3$. Give the interval of possible abscissae.

Worked answer: Applying and self-assessing

1. $\vec{v} = \frac{3}{2}\vec{u}$. 2. $B(4,5)$ and $C(7,7)$. 3. A direction vector is $(3,2)$; the equation is $2x - 3y + 7 = 0$, satisfied by the three points. 4. $|x-4| \leq 3 \iff x \in [1,7]$.

Applying and self-assessing

My priority:

The mistake to avoid:

The method to reuse:

An exercise to redo without help: